Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim:Perform FFT and IFFT on discrete time signal.**	
Experiment:-5	**Date:**	**Enrollment No:92400133074**

Github - https://github.com/pateldhananjay39/DSIP.git

Code:-

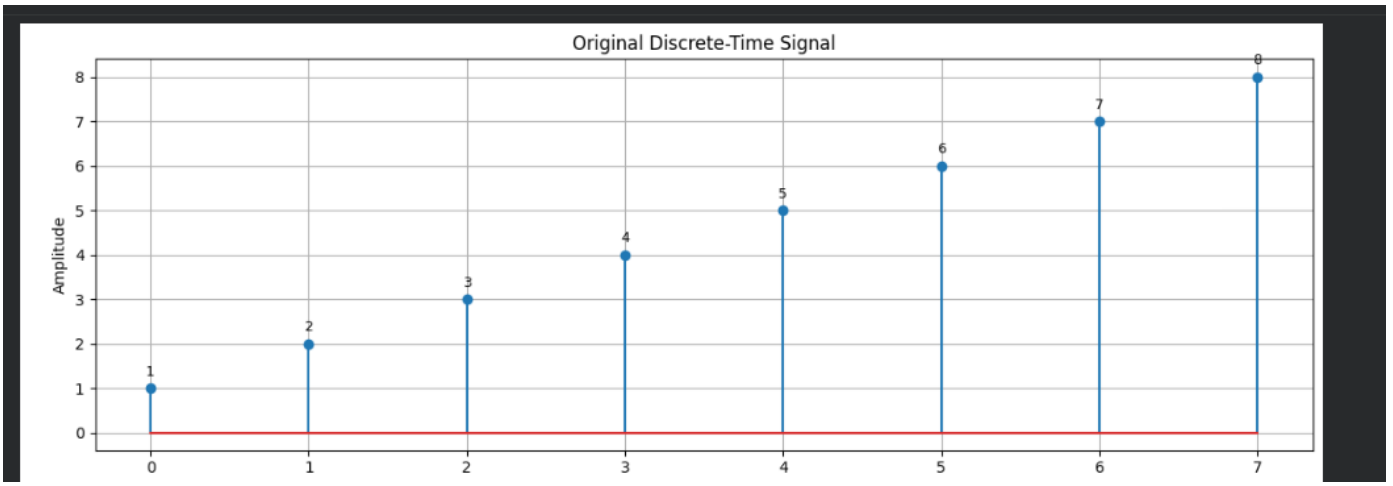
```
import numpy as np
import matplotlib.pyplot as plt
signal = np.array([1, 2, 3, 4, 5, 6, 7, 8])
# Compute the FFT of the signal
fft_result = np.fft.fft(signal)
# Compute the magnitude and Phase spectrum of the FFT result
magnitude_spectrum = np .abs(fft_result)
Phase_spectrum = np.angle(fft_result)
# Compute the IFFT of the FFT result
reconstructed_signal = np.fft.ifft(fft_result)
plt.figure(figsize=(12, 22)) # Adjust figure height for 5 subplots

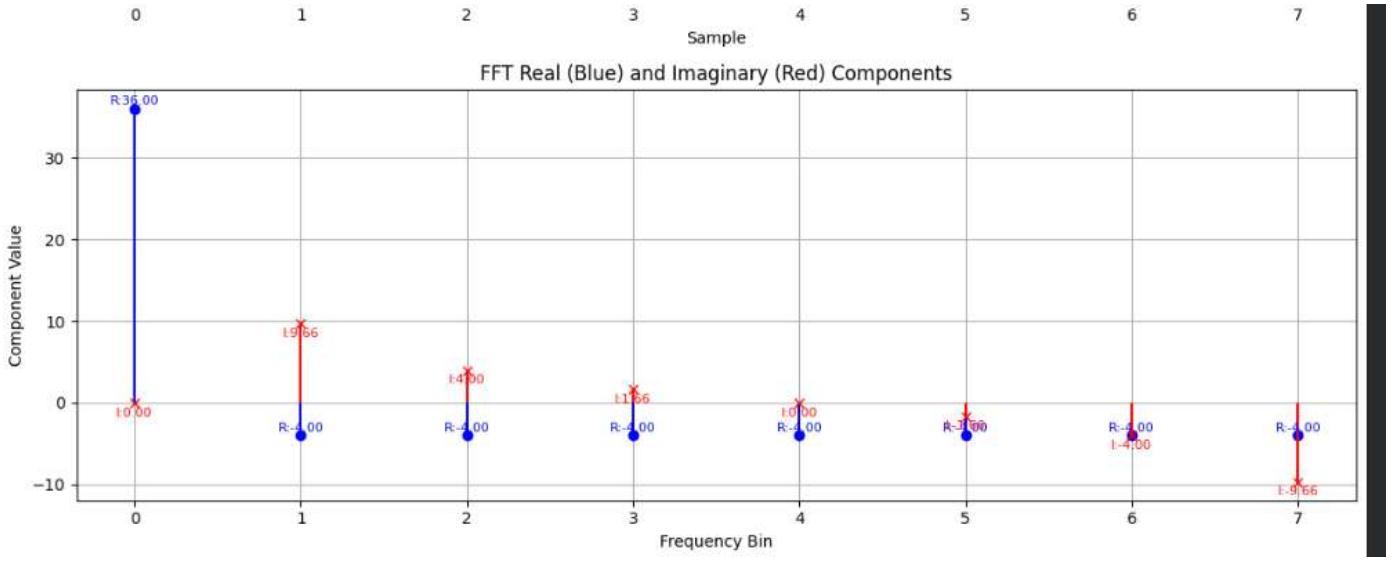
# 1. Original Discrete Time Signal
plt.subplot(5, 1, 1) # 5 rows, 1 column, 1st plot
plt.stem(signal)
plt.title('Original Discrete Time Signal')
plt.xlabel('Sample')
plt.ylabel('Amplitude')
plt.grid(True)
for i, val in enumerate(signal):
    plt.text(i, val + 0.2, f'{val}', ha='center', va='bottom', fontsize=9)
# 2. FFT Real and Imaginary Components
plt.subplot(5, 1, 2)
plt.stem(fft_result.real, linefmt='b-', markerfmt='bo', basefmt=' ')
plt.stem(fft_result.imag, linefmt='r-', markerfmt='rx', basefmt=' ')
plt.title('FFT Real (Blue) and Imaginary (Red) Components')
plt.xlabel('Frequency Bin')
plt.ylabel('Component Value')
plt.grid(True)
for i, val in enumerate(fft_result.real):
   plt.text(i, val + 0.2, f'R:{val:.2f}', ha='center', va='bottom', fontsize=9, color='blue')
for i, val in enumerate(fft_result.imag):
    plt.text(i, val - 0.5, f'I:{val:.2f}', ha='center', va='top', fontsize=9, color='red')
# 3. FFT Magnitude
plt.subplot(5, 1, 3)
plt.stem(magnitude_spectrum)
plt.title('FFT Magnitude')
```

```
plt.xlabel('Frequency Bin')
plt.ylabel('Magnitude')
plt.grid(True)
for i, val in enumerate(magnitude_spectrum):
    plt.text(i, val + 0.2, f'{val:.2f}', ha='center', va='bottom', fontsize=9)
# 4. FFT Phase Spectrum
plt.subplot(5, 1, 4)
plt.stem(Phase_spectrum)
plt.title('FFT Phase Spectrum')
plt.xlabel('Frequency Bin')
plt.ylabel('Phase (radians)')
plt.grid(True)
for i, val in enumerate(Phase_spectrum):
    plt.text(i, val + 0.1, f'{val:.2f}', ha='center', va='bottom', fontsize=9)
# 5. Reconstructed Signal
plt.subplot(5, 1, 5)
plt.stem(reconstructed_signal.real)
plt.title('Reconstructed Signal')
plt.xlabel('Sample')
plt.ylabel('Amplitude')
plt.grid(True)
for i, val in enumerate(reconstructed_signal.real):
    plt.text(i, val + 0.2, f'{val:.2f}', ha='center', va='bottom', fontsize=9)
plt.tight_layout()
plt.show()
```

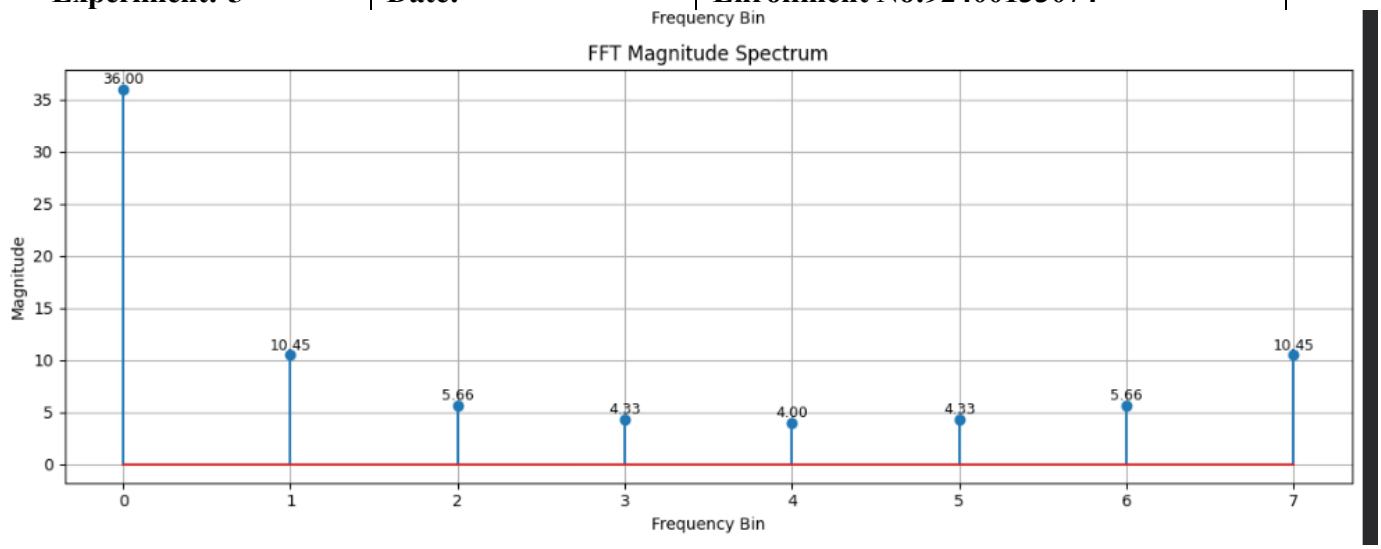
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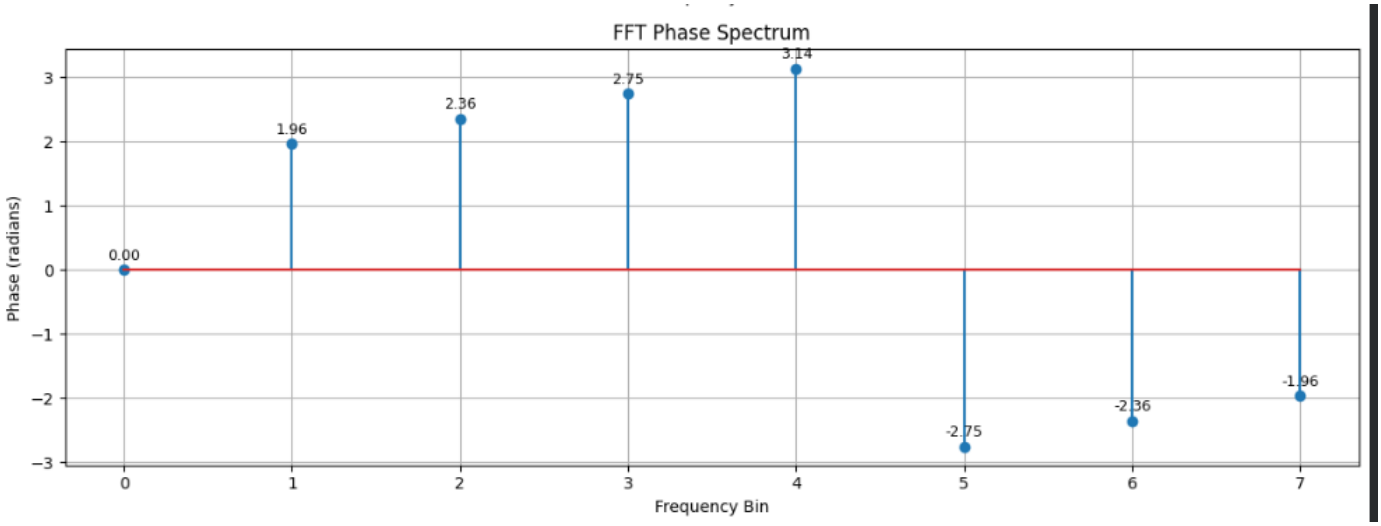
Output:-

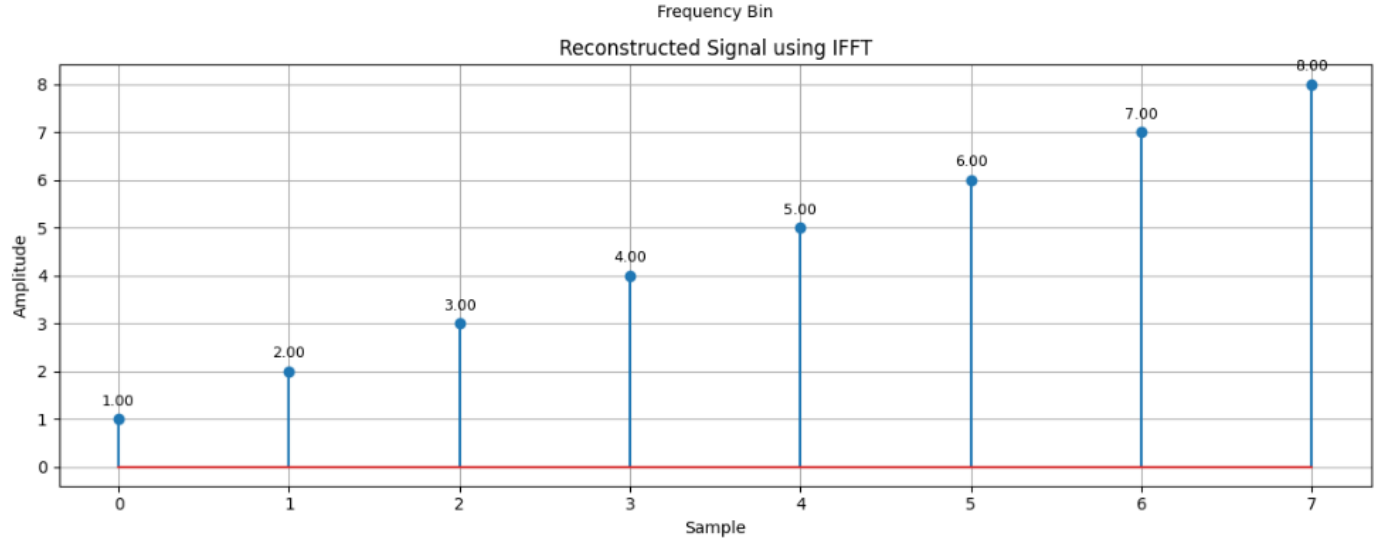




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```
iginal Signal: [1 2 3 4 5 6 7 8]
```

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Conclusion:

In this experiment, a discrete signal ranging from 1 to 8 was analyzed using the Fast Fourier Transform (FFT) to obtain its frequency components. The magnitude and phase spectra were examined to understand the contribution and phase of each frequency component. The original signal was then reconstructed using the Inverse Fast Fourier Transform (IFFT). The reconstructed output matched the original signal, confirming that FFT and IFFT provide an effective method for converting signals between the time and frequency domains and recovering the original data.